

EduGenie Proposed Solution

Project Overview

EduGenie is a lightweight web-based educational assistant. It uses FastAPI for the backend, HTML/CSS/JavaScript for the frontend, and Google Gemini for Generative AI capabilities.

Main Features

1. Intelligent Question Answering

The user enters a question and selects a learning level. EduGenie returns a direct answer with educational context.

2. Simplified Concept Explanation

The user enters a concept such as Pythagoras Theorem. EduGenie provides a definition, analogy, example, and key points.

3. AI-Powered Quiz Generation

The user selects a topic, difficulty, level, and question count. EduGenie creates a topic-specific quiz.

4. Educational Text Summarization

The user pastes study material. EduGenie creates a shorter summary while preserving important information.

5. Personalized Learning Roadmap

The user provides a topic, current level, available weekly study time, and learning goal. EduGenie creates a structured study plan.

Solution Modules

